

In VBA, **loops** are structures that allow you to repeat a block of code multiple times. They are indispensable for automating tasks across the **1,048,576 rows** available in an Excel worksheet, such as calculating every "Total" score in a student list or formatting a long "Salary statement".

1. For...Next Loop

This is the most common loop used when you know exactly how many times you want to repeat an action.

- **Syntax:**
- VBA

```
For i = 1 To 10  
    Cells(i, 1).Value = "Processed"  
Next i
```

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- **Best Use Case:** Iterating through a fixed range of cells, such as rows 1 to 100 in a specific column.
- **Step Keyword:** You can use `Step` to skip rows (e.g., `For i = 1 To 10 Step 2` will only process odd-numbered rows).

2. For Each...Next Loop

Instead of using a counter, this loop iterates through every object in a collection. It is often faster and more readable for Excel objects.

- **Syntax:**
- VBA

```
Dim cell As Range  
For Each cell In Range("A1:A10")  
    If cell.Value < 0 Then cell.Font.Color = vbRed  
Next cell
```

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 - **Best Use Case:** Applying **Conditional Formatting** logic or checking every worksheet in a workbook without needing to know the exact count.
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3. Do...While and Do...Until Loops

These loops continue as long as (or until) a specific condition is met. They are useful when you don't know the exact number of repetitions required.

- **Do While:** Repeats **while** a condition is True.
 - *Example:* Keep processing rows until you hit an empty cell in the "Student Name" column.
- **Do Until:** Repeats **until** a condition becomes True.
 - *Example:* Loop until the "Grand Total" reaches a specific target identified in a **What-If Analysis**.

4. Efficiency and Performance

When looping through large datasets, certain "hacks" prevent Excel from lagging:

- **Disable Screen Updating:** Use `Application.ScreenUpdating = False` at the start of your loop to stop the screen from flickering as it updates each cell.
- **Long Data Type:** Always declare your loop counters (like `i`) as a **Long** data type instead of an Integer to ensure the loop can handle more than 32,767 rows.
- **Exit Statement:** You can use `Exit For` or `Exit Do` to stop a loop immediately if a specific condition is met, saving processing time.

5. Comparison Table

Loop Type	Knowing the Count	Best For
For...Next	Yes	Specific ranges (e.g., Row 1 to 500).
For Each	No	Collections (e.g., all Sheets, all Charts).

Do While	No	Processing until data runs out.
Do Until	No	Reaching a specific numerical threshold.